AI and Data Science in Public Health

Graduate Seminar • 3 Credits • Fall 2025

Course Description

This course introduces students to machine learning and AI methods as applied to population health data. Topics include supervised and unsupervised learning, fairness and bias in predictive models, natural language processing for clinical text, and causal inference in observational studies. No prior programming experience required; Python is introduced from the ground up.

AI Use Policy

Students may use AI writing and coding assistants for exploration and drafting. All submitted work must include an "AI disclosure" section describing which tools were used and how. Final project code must be explainable by the student in a live code review. Undisclosed AI use is an academic integrity violation.

Schedule (selected weeks)

Week	Topic	Assignment due
1	Introduction: AI promises and limits in public health	—
3	Supervised learning: regression and classification	Problem set 1
5	Model evaluation and overfitting	Problem set 2
7	Fairness metrics and bias auditing	Midterm analysis
10	NLP for clinical notes	Problem set 3
13	Causal inference with ML	Problem set 4
15	Final project presentations	Final report

Grading

- Problem sets (4 × 15%): 60%
- Midterm analysis: 15%
- Final project: 25%